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FPGA Implementation of an ECG Arrhythmia Classifier

par *RTL Design, Fixed-Point Arithmetic, and On-Board Verification of the Pareto-03 1D-CNN on the Xilinx Virtex-6 ML605*

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Abstract. This report documents the hardware implementation of a compact 1D Convolutional Neural Network for real-time ECG arrhythmia classification on the **Xilinx Virtex-6 ML605 FPGA**. The architecture — selected from an NSGA-II Pareto front of 200 candidate models — contains only **6 868 parameters** with filter progression $[8, 32, 16, 16]$, achieving a Macro-F1 of 0.9887 at Float32 and 0.9800 after INT8 quantization. All weights are represented in **Q8.8 fixed-point** signed 16-bit format and stored in Xilinx Block RAMs initialised at bitstream load time via `$readmemh`. Custom parameterisable Verilog RTL modules were written for each layer. The complete system acquires ECG data from the AD8232 analog front-end, digitises it through the ADS1115 16-bit I^2C ADC, and streams 180-sample windows into the on-chip CNN engine. On-board verification uses 5 DIP switches to select among 30 stored test samples, with the classification result displayed on an HD44780 LCD.

Introduction

THE DEPLOYMENT of deep learning models on resource-constrained edge hardware demands a careful balance between classification accuracy, parameter count, and silicon area. This work focuses precisely on that intersection: implementing a Pareto-optimal neural network for cardiac arrhythmia detection directly on a programmable logic device.

The pipeline starts with signal acquisition from the patient's chest leads, passes through an analog conditioning stage and a 16-bit ADC, and ends with a real-time classification result on-chip. All stages from quantization strategy to Verilog module hierarchy are described in this report, with the intent of establishing a solid base for future tape-out using an open-source ASIC flow.

Machine Learning Foundation

The full ML pipeline is documented separately. This section summarises the key decisions that directly shape the hardware architecture.

Dataset & Pre-Processing

The MIT-BIH Arrhythmia Database provides 48 half-hour recordings sampled at $f_s = 360\,\text{Hz}$ from 47 subjects [1]. After a 51-tap FIR bandpass filter (f_L=0.5 Hz, f_H=45 Hz, Hamming window) and R-peak segmentation, each heartbeat is saved as a **180-sample Z-score normalised window**. Four arrhythmia classes are used:

ID	Class	Samples
0	Normal (N)	73 003
1	SVEB (Aberrant)	18 250
2	VEB (Bundle Branch)	18 250
3	Fusion / PVC	18 250
	Total	**127 753**

Table 1. Class distribution after noise-augmented balancing. Class weights: Normal 0.437, minority 1.75.

NSGA-II Architecture Search

Rather than designing the CNN by hand, a **multi-objective Neural Architecture Search** using NSGA-II [2] was executed over a search space of 38 880 candidate chromosomes. Each chromosome encodes the number of convolutional blocks (1–5) and per-block filter counts from $\{8, 16, 32, 64, 128, 256\}$. Two objectives are simultaneously minimised:

$$\min_{\mathbf{g}} \left[f_1(\mathbf{g}),\ f_2(\mathbf{g}) \right], \quad \begin{cases} f_1 = 1 - \text{Macro-F1} \\ f_2 = \text{parameter count} \end{cases} \tag{1}$$

The search runs G=10 generations with population N=20, 3-epoch proxy training per candidate, crossover p_c=0.9, and per-gene mutation p_m=0.2. The resulting Pareto front, illustrated in Figure 1, makes the accuracy–complexity trade-off explicit.

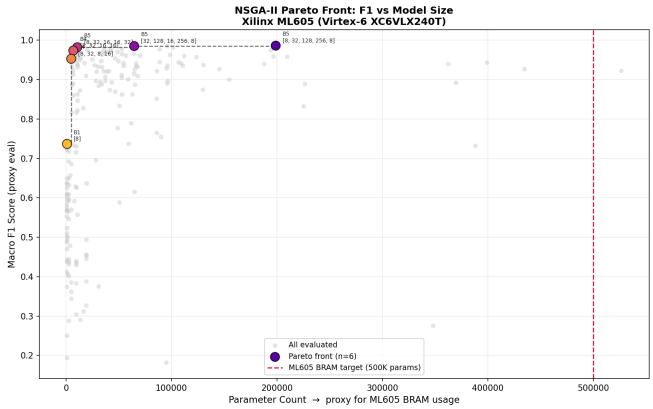

Figure 1. NSGA-II Pareto front. Each point represents a non-dominated architecture. Red dashed line: ML605 BRAM limit. The knee-point solution (Pareto-03) is highlighted: 4 blocks, filter progression $[8, 32, 16, 16]$, 6 868 parameters.

Why Pareto-03?

> **Selection rationale.** Pareto-03 sits at the knee of the Pareto curve — the point of maximum marginal gain. Any smaller model sacrifices significant accuracy; any larger model gains negligible accuracy at the cost of multiplied BRAM usage, pushing it out of the ML605 budget.

The selected architecture achieves **Macro-F1 = 0.9887** with only **6 868 parameters** — 98% fewer than a typical ResNet-based ECG classifier. The non-monotonic filter progression $[8, 32, 16, 16]$ is deliberate: Block 2 expands aggressively to capture diverse QRS morphologies; Blocks 3 and 4 compress back to reduce the BRAM footprint while retaining the encoded feature diversity.

INT8 Quantization

Post-training quantization via TensorFlow Lite converts all weights and activations to INT8 using the affine mapping:

$$x_q = \text{clip}\left(\text{round}\left(\tfrac{x}{S}\right) + Z,\ -128,\ 127\right) \tag{2}$$

Scale S and zero-point Z are calibrated from 100 representative training samples. The INT8 model occupies only **22 KB** on flash. Hardware simulation confirms a Macro-F1 of **0.9800** — a drop of only 0.87 percentage points from the Float32 baseline.

System Architecture

The full acquisition-to-classification chain is shown in Figure 2. The system is divided into three physical domains: analog acquisition, digital conversion, and on-chip inference.

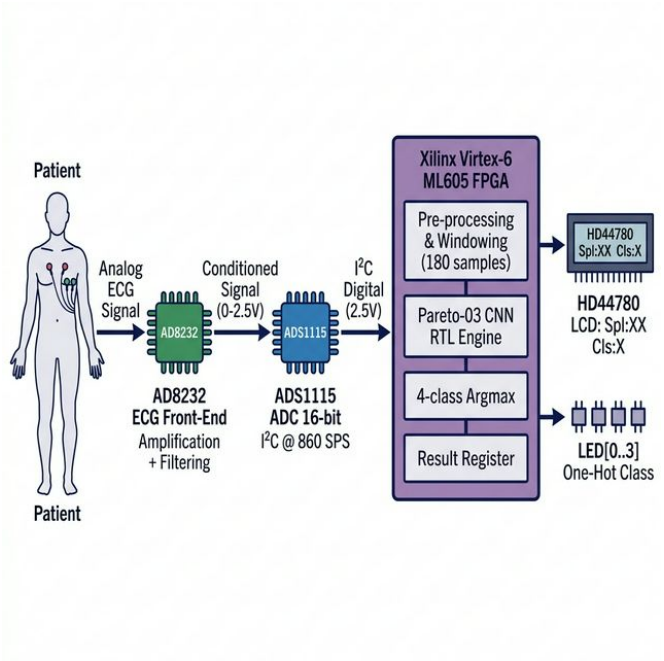

Figure 2. Complete end-to-end system architecture. ECG leads connect to the AD8232 analog front-end, which conditions the signal before digitisation by the ADS1115. Data enters the Virtex-6 via I^{2}C at 2.5 V, and the CNN result is displayed on the LCD and LEDs.

Analog Front-End: AD8232

The AD8232 is a single-lead ECG acquisition IC that integrates an instrumentation amplifier, a right-leg drive (RLD) for common-mode rejection, and a bandpass filter with configurable cutoff frequencies. It is powered at **2.5 V** — matching the FPGA I/O bank voltage — and outputs a conditioned analog signal centred at $V_{DD}/2 = 1.25\,\text{V}$.

Analog-to-Digital Converter: ADS1115

The ADS1115 is a 16-bit, 4-channel Σ–Δ ADC with an on-chip programmable gain amplifier (PGA) and I^{2}C interface. Its operating supply range of 2.0–5.5 V makes it directly compatible with the **2.5 V power rail** of the ML605 expansion headers, eliminating the need for a voltage level-shifter on the digital SDA/SCL lines.

> **Voltage compatibility.** By powering both the AD8232 and ADS1115 from the FPGA's 2.5 V rail, the I^{2}C bus swings at 2.5 V natively. No level shifter is required, and the ADC analog input remains within $V_{DD} + 0.3 = 2.8\,\text{V}$ absolute maximum throughout the ECG waveform range.

The ADS1115 is configured for **860 SPS** (samples per second) continuous conversion on channel AIN0. At $f_s = 360\,\text{Hz}$ (MIT-BIH original), a 180-sample window covers one complete cardiac cycle (approximately 500 ms). The I^{2}C master on the FPGA reads one conversion result per I^{2}C transaction.

FPGA Target: Xilinx Virtex-6 ML605

The ML605 evaluation board hosts a Virtex-6 XC6VLX240T-1FF1156, providing:

Resource	Available
Logic Slices	37 680
DSP48E1	768
Block RAM (36 Kb)	416
I/O Banks	12 (2.5 V on Banks 13, 14)

Table 2. Virtex-6 XC6VLX240T resource summary.

The CNN inference engine uses only **4 DSP48E1 slices and 1 BRAM36**, representing less than 0.6% and 0.2% utilisation respectively.

RTL Implementation

Figure 3 shows the complete RTL pipeline. All modules are written in structural Verilog with explicit parameter passing for filter counts and fixed-point widths.

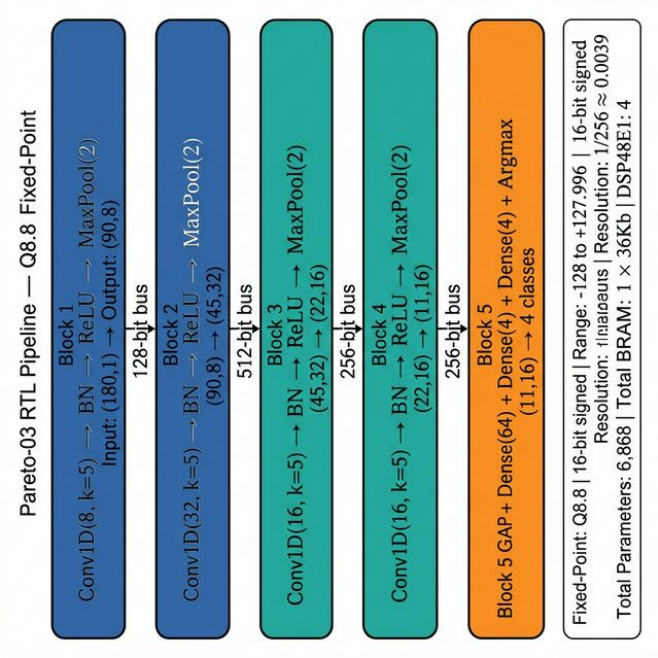

Figure 3. Pareto-03 RTL pipeline. Each block processes a stream of 16-bit Q8.8 samples. Internal bus widths scale with the number of active filter channels.

Fixed-Point Format: Q8.8

All weights, activations, and intermediate results use **Q8.8 signed 16-bit two's complement**:

$$v_{\text{real}} = \frac{v_{\text{hex}}}{256}, \quad \text{Range: } -128.0 \le v \le +127.996 \tag{3}$$

The resolution of $1/256 \approx 0.0039$ is sufficient to represent the INT8-quantized weights with full fidelity. When two Q8.8 values are multiplied, the result is Q16.16 (32-bit), preserving precision before the Batch Normalization rescaling step clips it back to Q8.8.

Convolutional Layer: `layer_conv_k5`

A sliding 5-tap FIR-style window accumulates the MAC output at every clock cycle. The module is parameterised by `FILTER_ID` so that multiple instances can be generated in a loop using Verilog `generate`:

Conv1D module: Q8.8 MAC

```
always @(*) begin
  sum = bias_shifted; // Q8.8 -> Q16.16
  for (m = 0; m < 5; m = m + 1)
    sum = sum + (window[4-m] * weights[FILTER_ID*5 +
      m]);
end
// MSB[31] = sign -> ReLU mask
```

Output precision is 32 bits (Q16.16) to prevent overflow before Batch Normalization. The `STREAM_LENGTH` parameter triggers a flush of the last $\lfloor k/2 \rfloor$ samples after the final input to complete the "same" padding convolution.

Batch Normalization: `layer_batchnorm`

The standard inference-time formula involves division and a square root — both expensive on an FPGA. These are folded offline into two static coefficients before synthesis:

$$W_{bn} = \frac{\gamma}{\sqrt{\sigma^2 + \varepsilon}}, \quad B_{bn} = \beta - \mu W_{bn} \tag{4}$$

At runtime, the FPGA executes only a single MAC: $y = (x \times W_{bn}) + B_{bn}$. The ReLU activation is implemented as a one-bit sign check: if `sum[31] == 1` (negative), the output is clamped to zero.

MaxPooling: `layer_maxpool`

A two-register sliding comparator advances by one position per clock and emits the maximum of every pair of consecutive values, halving the time-step dimension with a stride of 2.

Dense Layers: `layer_dense0` / `layer_dense1_out`

The fully connected layers perform a combinational MAC across all input channels simultaneously. Since the input to `layer_dense0` is only 16 values wide (GAP output), and `layer_dense1` is 64 wide, XST infers distributed LUT RAM rather than Block RAM. This keeps the BRAM budget near zero while the combinational latency remains acceptable at 50 MHz.

Global Average Pooling: `layer_gap`

The GAP module accumulates all T time-step values for each channel over T clock cycles, then divides by T using a right-shift approximation ($T = 11 \approx 2^{3.46}$; division by 11 is pre-calculated and implemented as a fixed multiply-shift chain). The output is a 16-element vector fed directly to `layer_dense0`.

Weight Initialisation via `$readmemh`

All weight and bias tensors are loaded at power-on from `.hex` files using Verilog `initial` blocks. In ISE 14.7, XST embeds

these hex initialisation values directly into the BRAM `INIT` attributes of the generated bitstream — the weights are physically baked into the configuration file and require no external memory at runtime.

ISE Synthesis Setup

The design targets **Xilinx ISE 14.7** with the XST synthesis engine. The project is structured under `Synthesis/` with a dedicated `ise/` subdirectory holding the top-level and constraint files.

Clocking Strategy

The ML605 provides a 200 MHz LVDS differential clock on pins J9 (P) and H9 (N), Bank 13. A **Clocking Wizard** IP (CORE Generator) divides this to **50 MHz** for the CNN datapath and LCD controller. The differential input is buffered internally by the wizard's `IBUFGDS` primitive. The reset is synchronised to the 50 MHz domain and held active until the wizard's `LOCKED` signal is asserted.

Pin Constraints (UCF)

All pin assignments were verified against the ML605 Hardware User Guide UG534. The complete set of used I/O is:

Signal	LOC	Standard	Function
`sys_clk_p/n`	J9/H9	LVDS_25	200 MHz input
`reset`	G26	LVCMOS25	Centre push button
`led[0..3]`	AC22..AE23	LVCMOS25	One-hot class
`sw[0..4]`	D22..C18	LVCMOS25	DIP sample select
`LCD_*`	Bank 13	LVCMOS25	HD44780 4-bit

Table 3. UCF pin assignments (UG534-verified).

On-Board Test Infrastructure

To validate inference without a live ADC connection, 30 pre-digitised ECG test samples are stored in a single Block RAM ROM (`test_all_samples_rom`), initialised from `combined_test_inputs.hex` (5 400 × 16-bit words). The five **User DIP Switches** (DIP0–DIP4, pins D22–C18) encode a 5-bit binary index (0–29) that selects the base address:

$$\text{addr_base} = \text{sw}[4:0] \times 180 \tag{5}$$

After a switch change or power-on reset, the ROM streams 180 samples into the CNN at one sample per clock cycle. The classification result is latched and displayed on:

- **LEDs**: LED[0..3] one-hot encoding of the predicted class.
- **LCD**: Line 1 shows sample index and class index (`Spl:XX Cls:Y`); Line 2 shows the class name (`NORMAL`, `SVEB`, `VEB`, `FUSION`).

Resource Utilisation

Resource	Used	Budget	Status
DSP48E1	4	768	✓ 0.5%
BRAM36	1	416	✓ 0.2%
INT8 Params	6 868	500K	✓ 1.4%
Model Size	22 KB	3.7 MB	✓ 0.6%

Table 4. Estimated Virtex-6 resource usage for the Pareto-03 CNN (from `ml605_resource_report.txt`).

The remarkably low DSP usage is a direct consequence of the NSGA-II size objective: the search penalised large convolutional kernels and wide filter banks. The entire parameter set fits within a single BRAM36 tile with room to spare.

Future Work: ASIC Physical Design

Planned extension. The natural next step after FPGA validation is to port the RTL to a physical silicon implementation using an open-source ASIC flow.

The FPGA implementation establishes the RTL correctness of the Pareto-03 engine. The logical progression toward a silicon realisation follows the standard ASIC design flow: RTL synthesis → floorplanning → placement → clock tree synthesis → routing → sign-off. This work plans to target the **SkyWater SKY130 130 nm PDK** through the **OpenROAD / OpenLane 2** open-source EDA toolchain.

OpenLane 2 & SKY130

OpenLane 2 is an automated RTL-to-GDS flow built around OpenROAD, Yosys, and Magic. The SKY130 process design kit provides standard cell libraries at 130 nm feature size with five metal layers. At 130 nm, a design of 6 868 18-bit registers occupies roughly 0.05 mm^2 in core area — feasible for a first educational tape-out through programs such as **Google/efabless chipIgnite** or the **IHP SG13G2** (130 nm SiGe BiCMOS).

Bridge from FPGA to ASIC

The Static Timing Analysis (STA) infrastructure already established in `pareto_03FPGAsim/STA/` provides the timing constraints that will be adapted as SDC files for the ASIC flow. The XDC/UCF constraints are converted to SDC format:

```
create_clock -period 20.0 [get_ports clk]   (50 MHz)
```
(6)

Key FPGA-to-ASIC differences to address include replacing Block RAM with synthesised SRAM macros, adapting the I/O ring for wire bonding, and running DRC/LVS verification with Calibre or Magic/Netgen.

References

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